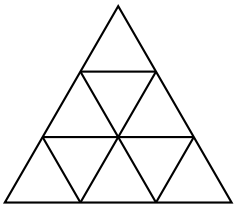
Combinatorics Olympiad
Advanced & Free Levels
October 2025

Problems

1. Nine **pairwise distinct** positive integers are written in each of the 9 cells of the following figure. Parallel to each side of the outer triangle, there are three rows that contain 1, 3, and 5 cells, respectively. This defines a total of 9 distinct subsets of cells (3 rows in each of the 3 directions). Suppose that among these 9 subsets, there are k subsets whose sums of numbers are equal. What is the largest possible value of k, over all possible assignments of the initial 9 numbers to the cells?



2. An $m \times n$ table is partitioned along its grid lines into several pieces (not necessarily rectangles). The pieces are then reassembled, without rotation or reflection and only through translation, to form a $n \times m$ table. Prove that this can be accomplished by partitioning the original table into at most $|m - n| + 1$ pieces.

3. A table is tiled with black tiles of sizes $1 \times 1, 1 \times 2, 1 \times 3, 1 \times 4$, and 1×5 (each tile may have appeared horizontally or vertically). Is it always possible to color exactly one square of each tile white, such that a chess king placed on any white square can exit the table by only moving to other white squares?

4. 100 people are working in a company. We know that at least 1 and at most 98 of them are thieves. This company has m warehouses, and each of the 100 people has been told the location of some non-zero number of these warehouses. On the first day of the year, all thieves report the locations of all the warehouses that they know to their chief (who is out of the company), and their chief attacks all the reported warehouses. Fortunately, the way the people were informed of the warehouses' locations guarantees that after this attack, we can identify at least one of the thieves. Find the smallest value of m that satisfies this condition.

5. Let S be a set of n points in the plane. For each labeling L of the points in S with numbers from 1 to n, a graph $G(L)$ with the vertex set $\{v_1, v_2, \ldots, v_n\}$ is defined as follows:

   > For each i from 2 to n, we connect vertex v_i to the vertex v_j where j is the label of the point which is the farthest point from point i among the points with labels $\{1, 2, \ldots, i-1\}$; If there are more than one such point in this set, the one with the **smallest** label is chosen.

   Let $f(L)$ be the maximum degree of the vertices in the graph $G(L)$, and let $h(S)$ be the minimum value of $f(L)$ over all $n!$ possible labelings of the members of S.

   (a) Provide a set S such that $h(S) = 3$.

   (b) For any set S, prove that there exists a finite set of points T, such that $S \subseteq T$ and $h(T) = 2$.

6. Dara and Sara are playing a game. Assume that at the beginning of the game, there are n containers, each with some number of stones. The operation "throw" consists of choosing a container and throwing a stone into it. The game starts with Dara, and they take turns alternatively. Dara makes one throw in each of his turns, while Sara makes $n-1$ throws in each of her turns. Prove that Sara can play in such a way that, at least once, at the end of some player's turn, the greatest common divisor of the numbers of stones in the containers becomes greater than 1.

7. In a gathering of 2025 people, a game is to be played. Each person, independent of the others, chooses another person from the group to be his "slap target". Each person is unaware of who others have chosen. After the choices are made, a random order of these individuals is selected. The people, in the specified order, insert themselves into a queue. The i-th person to enter has i choices for his position in the queue among the previous people. Each person's goal is to end up directly behind his slap target. Therefore, each person, among all possible positions, chooses the one that maximizes his probability of achieving his goal. Furthermore, each person knows that each of the other people is also pursuing his own goal. What is the probability that a given person will achieve his goal?